CONFIDENTIAL **MOCK EXAM 2026**

GENERIC UNIVERSITY FINAL EXAMINATION

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of two (2) parts: PART A (10 Questions)
   PART B (2 Questions)

2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.

3. Do not bring any material into the examination room unless permission is given by the invigilator.

4. Please check to make sure that this examination pack consists of:

   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty

5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 4 printed pages

PART A (80 MARKS)

QUESTION 1

a) Explain where network-based IDS sensors should be placed in a network and why their placement matters.

(2 marks)

b) Describe six preventive network security mechanisms and, for each, state the specific protective role it plays in an organization's network.

(6 marks)

QUESTION 2

a) List TWO (2) reasons why software is insecure, as discussed in the Software Security lecture.

(2 marks)

b) State six (6) Windows security mechanisms or components, as discussed in the Windows Security lecture, and briefly indicate the role of each.

(6 marks)

QUESTION 3

a) State TWO (2) characteristics of a Trojan horse as a type of malware.

(2 marks)

b) Discuss six (6) ways in which malware attacks and propagates, referring to historical malware examples, executable infection techniques, and the behaviour of rootkits and spyware.

(6 marks)

QUESTION 4

a) Explain the Chinese Wall model as a formal security model, stating the concern it is designed to address and how its access decisions differ from static policies.

(2 marks)

b) Explain the three main access control structures used to implement access policies, namely the access control matrix, access control lists and capabilities, and state the requirements that any such structure must satisfy.

(6 marks)

QUESTION 5

a) List two goals of access control and state the instrument used to make authorization decisions.

(2 marks)

a) Refer to your answer in (a). Discuss six distinct features of formal security models, covering the Bell-La Padula, Biba and Chinese Wall models, including what each model controls and how the Chinese Wall model's rules constrain read and write access.

(6 marks)

QUESTION 6

a) Explain how a disclosure threat and a usurpation threat each cause harm to a system, stating which security objective is affected in each case.

(2 marks)

b) The Samy worm spread rapidly across MySpace by exploiting a cross-site scripting flaw. Explain six (6) aspects of this incident, covering the underlying XSS vulnerability, how the worm injected and propagated itself, and the scale of the resulting infection.

(6 marks)

QUESTION 7

a) Explain how the chain of trust works in Trusted Computing, referring to the role of the TPM and the stages that are measured and validated.

(2 marks)

b) Describe six (6) components and characteristics of the NGSCB/Nexus architecture, covering both the Nexus kernel and the Nexus Computing Agent, as presented in the Trusted Computing lecture.

(6 marks)

QUESTION 8

a) State TWO (2) differences between morals and laws as forms of conduct guidance for computing professionals.

(2 marks)

b) Discuss how a professional code of ethics and conduct, such as the ACM code, guides a computing professional in carrying out security-related work. Your discussion should cover six (6) distinct obligations or expectations that such a code places on the professional.

(6 marks)

QUESTION 9

a) Explain the stages involved in the biometric authentication process, distinguishing clearly between the identification stage and the authentication stage.

(2 marks)

b) Biometric technologies are grouped into different categories and each technology has its own operational strengths and weaknesses. Explain the categories of biometric applications and the characteristics of specific biometric technologies, giving six distinct points.

(6 marks)

QUESTION 10

a) State TWO (2) limitations of misuse detection systems when used to detect attacks.

(2 marks)

b) State the distinction between a vulnerability, a threat and an attack as presented in the security fundamentals lecture, and state one countermeasure principle for protecting assets against such dangers.

(6 marks)

PART B (20 MARKS)

QUESTION 1

a) State FIVE (5) password security controls that can be applied to protect a system's password files and user accounts.

(5 marks)

b) Explain the design criteria that Kerberos satisfies and the benefits that motivate organisations to adopt single sign-on solutions such as Kerberos, smart cards and two-factor authentication.

(5 marks)

QUESTION 2

a) [THEORY PLACEHOLDER - risk/management (ethics), 5 marks, Explain. No sourced question passed verification.]

(5 marks)

b) [THEORY PLACEHOLDER - risk/management (ethics), 5 marks, Discuss. No sourced question passed verification.]

(5 marks)

END OF QUESTION PAPER